Steering control for control-affine systems via nonlinear controllability Gramian

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Association Science pour l'Afrique (ASA) Seminar

March 8, 2026

Outline

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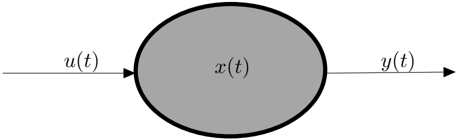
Notion of control systems

A control system in this talk is a **dynamical system** on which one can act by using suitable **controls**.



Mathematically, it often takes the form

$$\dot{x}(t) = f(t, x(t), u(t)), \qquad y(t) = g(t, x(t), u(t)). \qquad (\Sigma)$$

- $x(\cdot)$ is the **state** of the system and $u(\cdot)$ is the **control**;
- $y(\cdot)$ is the **output**, i.e., represents what we **observe**/**measure** from the system;
- A solution $t \in [t_0, T] \mapsto (x(t), u(t))$ of $\dot{x}(t) = f(t, x(t), u(t))$ is called a **trajectory** of (Σ);
- $\Omega \subseteq \mathbb{R}^d$ is the **state space** and $\mathcal{U} \subseteq L^\infty((t_0, T); \mathbb{R}^k)$ is the set of **admissible controls**; $k \leq d$.

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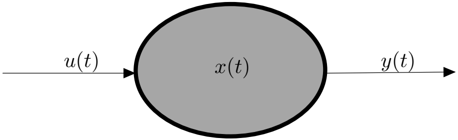

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Control-affine systems: reachable set & talk's focus

$$(\Sigma): \begin{cases} \dot{x}(t) = N(x(t)) + B(x(t))u(t) \\ x(0) = x^0 \in \Omega \end{cases}$$

Reachability

For $x^0 \in \Omega$, we say that $x^1 \in \Omega$ is reachable on $[0,T]$ from x^0 if $\exists\, u \in L^\infty((0,T);\mathbb{R}^k)$ s.t. the solution $x_u(\cdot)$ of (Σ) satisfies $x_u(T) = x^1$.

Reachable set: formal definition

The reachable set is defined by

$$\mathcal{R}_\Sigma(T, x^0) := \left\{ x_u(T) : \begin{array}{l} x_u(\cdot) \text{ solution of } (\Sigma), \\ x(0) = x^0 \end{array} \right\}.$$

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Complete controllability

(Σ) is said to be completely controllable (or controllable) on $[0, T]$ for some $T > 0$ if $\mathcal{R}_\Sigma(T, x^0) = \Omega$ for all $x^0 \in \Omega$.

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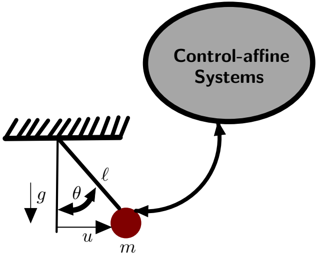
Objective of the talk

- **Control analysis:** characterize $\mathcal{R}_\Sigma(T,x^0)$ or provide a *quantitative subset*;
- **Control synthesis:** given $x^1 \in \mathcal{R}_\Sigma(T,x^0)$, synthesize a steering control from x^0 to x^1;
- **Minimum-energy control:** among steering controls from x^0 to $x^1 \in \mathcal{R}_\Sigma(T,x^0)$, find one with minimum L^2-norm.

Examples of control-affine systems



Torque-controlled pendulum

$$\dot{\theta} = \omega, \quad \dot{\omega} = -a \sin\theta + bu$$
$$a := g/\ell, \quad b := 1/m\ell^2.$$

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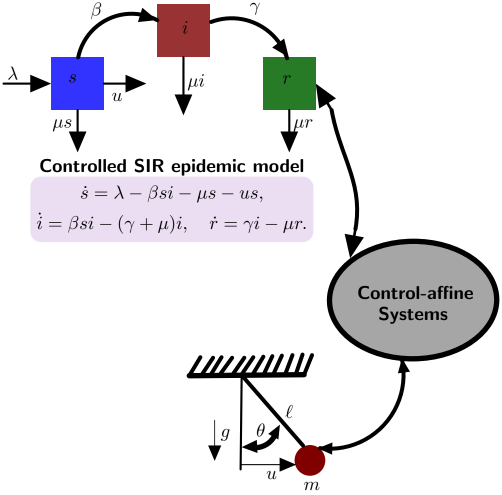

Controlled SIR epidemic model

$$\dot{s} = \lambda - \beta s i - \mu s - u s,$$
$$\dot{i} = \beta s i - (\gamma + \mu) i, \quad \dot{r} = \gamma i - \mu r.$$

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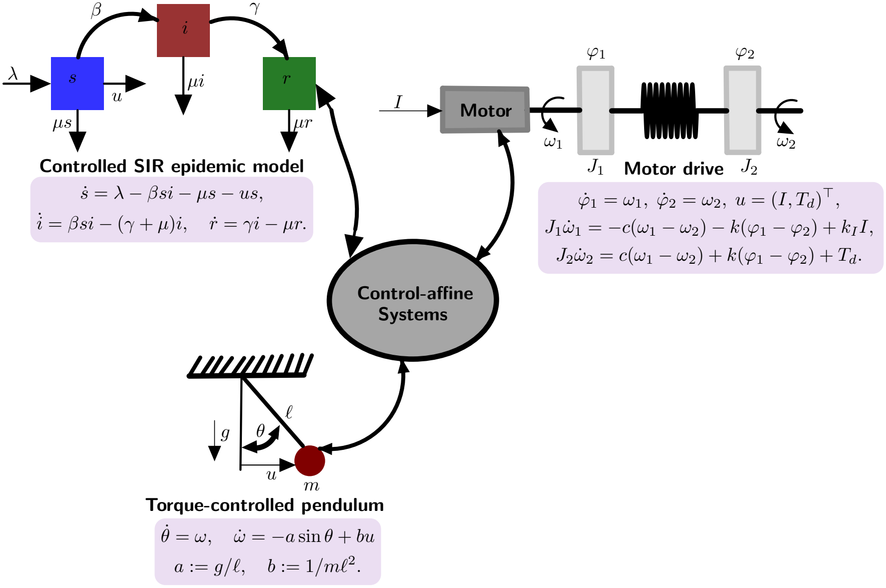

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Motor drive

$$\dot{\varphi}_1 = \omega_1, \ \dot{\varphi}_2 = \omega_2, \ u = (I, T_d)^\top,$$
$$J_1\dot{\omega}_1 = -c(\omega_1 - \omega_2) - k(\varphi_1 - \varphi_2) + k_I I,$$
$$J_2\dot{\omega}_2 = c(\omega_1 - \omega_2) + k(\varphi_1 - \varphi_2) + T_d.$$

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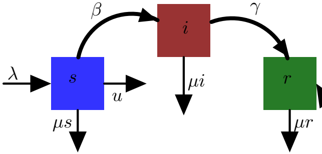

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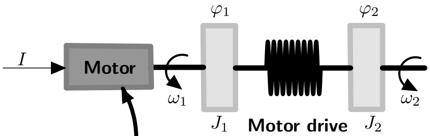

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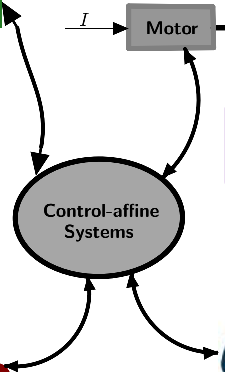

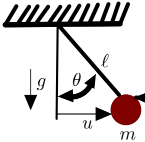

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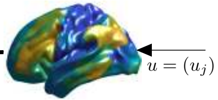

Brain-controlled dynamics

$$\dot{z}_i = -d_i\, z_i + \sum_{j=1}^{d} w_{ij}\sigma_j(z_j) + \sum_{j=1}^{k} b_{ij}u_j, \qquad i = 1, \cdots, d.$$

Outline

Linear time-invariant systems [Kalman, early 60]

$$(\Sigma): \begin{cases} \dot{x}(t) = Ax(t) + Bu(t) \\ x(0) = x^0 \in \mathbb{R}^d \end{cases}$$

Control analysis

1. $A \in \mathbb{R}^{d\times d}$, $B \in \mathbb{R}^{d\times k}$, $x(\cdot) \in \mathbb{R}^d$, and $u(\cdot) \in \mathbb{R}^k$;
2. $\forall x^0 \in \mathbb{R}^d$, $\mathcal{R}_\Sigma(T, x^0) = e^{TA}x^0 + \mathrm{Range}(W_c)$ where $W_c := \displaystyle\int_0^T e^{tA}BB^\top e^{tA^\top}\, dt$;
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Control synthesis & Minimum-energy control

1. The control $u(t) = B^\top e^{(T-t)A^\top} W_c^{-1}(x^1 - e^{TA}x^0)$ steers (Σ) from x^0 to $x^1 \in \mathcal{R}_\Sigma(T, x^0)$ in time T;
2. u is the minimum-energy control among all controls steering any $x^0 \in \mathbb{R}^d$ to $x^1 \in \mathcal{R}_\Sigma(T, x^0)$.

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1. Are based on the variation of constants formula: $\forall t \in [0, T]$,
$$x(t) = e^{tA}x^0 + \int_0^t e^{(t-s)A}Bu(s)\, ds;$$
2. Uniform w.r.t. $T > 0$ and $x^0 \in \mathbb{R}^d$: If $\mathcal{R}_\Sigma(T_0, 0) = \mathbb{R}^d$ for some $T_0 > 0$, then $\mathcal{R}_\Sigma(T, x^0) = \mathbb{R}^d$, $\forall T > 0$ and $\forall x^0 \in \mathbb{R}^d$;
3. Does not account for nonlinearities in $x(\cdot)$ and/or $u(\cdot)$.

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Frameworks, flow maps of the drift

$$(\Sigma) : \begin{cases} \dot{x}(t) = N(x(t)) + B(x(t))u(t) \\ x(0) = x^0 \in \mathbb{R}^d \end{cases}$$

Assumption on the drift term

1. $N \in C^2(\mathbb{R}^d)$ is globally Lipschitz and $\|DN(x)\| \leq \Lambda_1$, $\forall x \in \mathbb{R}^d$;
2. $\|D^2N(x)\| \leq \Lambda_2$, $\forall x \in \mathbb{R}^d$.

Assumption on the input matrix & control

1. $B \in L^\infty(\mathbb{R}^d; \mathbb{R}^{d\times k})$, $x \mapsto B(x)$ is $C^1(\mathbb{R}^d)$ and globally Lipschitz;
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- $(\phi_t)_{t\in\mathbb{R}}$ flow of N: $\partial_t\phi_t(x) = N(\phi_t(x))$, $\phi_0(x) = x$;
- The Jacobian matrix $D\phi_t(x)$ satisfies: $\partial_t D\phi_t(x) = DN(\phi_t(x))D\phi_t(x)$, $D\phi_0(x) = \mathrm{Id}$.

$\forall \alpha_t \in C^1(\mathbb{R}_+; \mathbb{R})$, $\alpha_0 \geq 0$, $\forall \beta_t \in C^0(\mathbb{R}_+; \mathbb{R}^d)$, $\forall t \geq 0$:

- $\|D\phi_{\alpha_t}(\beta_t)\| \leq e^{\Lambda_1\alpha_0}e^{\Lambda_1\int_0^t|\dot{\alpha}(\tau)|d\tau}$;
- $\|D^2\phi_{\alpha_t}(\beta_t)\| \leq \Lambda_2 e^{2\Lambda_1\alpha_0}e^{\Lambda_1\int_0^t|\dot{\alpha}_\tau|d\tau}(\alpha_0+e^{\Lambda_1\int_0^t|\dot{\alpha}_\tau|d\tau}\int_0^t|\dot{\alpha}_\tau|d\tau)$.

Solution representation & Gramian maps

$$(\Sigma): \begin{cases} \dot{x}(t) = N(x(t)) + B(x(t))u(t) \\ x(0) = x^0 \in \mathbb{R}^d \end{cases}$$

Theorem 1 [T, Ching; '25], [Agrachev, Gamkrelidze; '79]

For all $x^0 \in \mathbb{R}^d$ and $u \in L^\infty := L^\infty((0,T);\mathbb{R}^k)$ the solution $x_u(\cdot) \in C^0([0,T];\mathbb{R}^d)$ of (Σ) expresses for $t \in [0,T]$ as

$$x_u(t) = \phi_{t-T}\left(\phi_T(x^0) + \int_0^t D\phi_{T-s}(x_u(s))\, B(x_u(s))\, u(s)\, ds\right).$$

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Definition [T, Ching; '25]

Fix $x^0 \in \mathbb{R}^d$. The **Gramian map** $\mathcal{N} : L^\infty \to \mathcal{S}_d^+(\mathbb{R})$ is defined for every $u \in L^\infty$ and $x_u(\cdot)$ solution of (Σ) by:

$$\mathcal{N}(u) = \int_0^T D\phi_{T-t}(x_u(t))B(x_u(t))B(x_u(t))^\top D\phi_{T-t}(x_u(t))^\top\, dt.$$

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If $N(x) = Ax$ and $B(x) = B$, then

1. $x_u(t) = e^{tA}x^0 + \int_0^t e^{(t-s)A}Bu(s)\, ds;$
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A priori analysis

1. From $x_u(t) = \phi_{t-T}\left(\phi_T(x^0) + \int_0^t D\phi_{T-s}(x_u(s))\, B(x_u(s))\, u(s)\, ds\right)$, one has $x_u(T) = x^1$, iff

$$\int_0^T D\phi_{T-t}(x_u(t))\, B(x_u(t))\, u(t)\, dt = x^1 - \phi_T(x^0). \tag{1}$$

2. Fix $u \in L^\infty$, define the bounded linear operator $\mathcal{L}_u : L^2 := L^2((0,T);\mathbb{R}^k) \to \mathbb{R}^d$ by

$$\mathcal{L}_u v = \int_0^T D\phi_{T-t}(x_u(t))\, B(x_u(t))\, v(t)\, dt \qquad v \in L^2. \tag{2}$$

3. If for some $u \in L^\infty$, $y := x^1 - \phi_T(x^0) \in \mathrm{Range}(\mathcal{L}_u)$, then $\min\{\|v\|_{L^2} : \mathcal{L}_u v = y\}$ has the solution

$$v(u)(t) := (\mathcal{L}_u^\dagger y)(t) = B(x_u(t))^\top D\phi_{T-t}(x_u(t))^\top \mathcal{N}(u)^\dagger y \quad t \in [0,T]. \tag{3}$$

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Feasible coercivity class

Key remark

If one can prove for some $u \in L^\infty$ that $\mathcal{N}(u)$ is invertible and u satisfies the implicit relation

$$u(t) = B(x_u(t))^\top D\phi_{T-t}(x_u(t))^\top \mathcal{N}(u)^{-1}\left(x^1 - \phi_T(x^0)\right) \qquad t \in [0,T], \tag{4}$$

then the solution $x_u(\cdot)$ of (Σ) with control $u(\cdot)$ satisfies $x_u(T) = x^1$.

Definition [T, Ching; '25]

Let $C := C(\|B\|_\infty, \Lambda_1, \Lambda_2, T$, and possibly $x^0) > 0$, we define the *feasible coercivity class* by

$$\mathcal{F}(C) := \left\{u \in L^\infty((0,T);\mathbb{R}^k) : \ \lambda_{\min}\big(\mathcal{N}(u)\big) \geq C^{-1}\right\}. \tag{5}$$

- *Possibility of emptiness:* For a given $C > 0$, $\mathcal{F}(C) = \emptyset$ is possible;
- *Monotonicity:* If $C \leq C'$ then $\mathcal{F}(C) \subset \mathcal{F}(C')$;
- *Topological property:* $\mathcal{F}(C)$ is closed in L^∞ since $u \mapsto \lambda_{\min}(\mathcal{N}(u))$ is continuous.

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Synthesis maps & feasible coercivity ball

Definition [T, Ching; '25]

- *Feasible coercivity class:* $\mathcal{F}(C) := \Big\{u \in L^\infty((0,T);\mathbb{R}^k) : \ \lambda_{\min}\big(\mathcal{N}(u)\big) \geq C^{-1}\Big\}$;
- *Synthesis map.* $\mathcal{S} : \mathcal{F}(C) \to L^\infty$ by

$$\mathcal{S}(u)(t) = B(x_u(t))^\top D\phi_{T-t}(x_u(t))^\top \mathcal{N}(u)^{-1}\, y \qquad u \in \mathcal{F}(C),\ t \in [0,T]. \tag{6}$$

- *A priori* estimate on $\mathcal{F}(C)$:

$$\|\mathcal{S}(u)\|_\infty \leq \zeta(y) := C\,\|B\|_\infty\, e^{\Lambda_1 T}\,|y| \qquad \forall u \in \mathcal{F}(C). \tag{7}$$

- *Feasible coercivity ball:*

$$\mathcal{F}(y) := \{u \in \mathcal{F}(C) : \|u\|_\infty \leq \zeta := \zeta(y)\}. \tag{8}$$

Fixed point of the synthesis map via Cacciopolli fixed-point theorem

Theorem 2 [T, Ching; '25]

Assume that $\mathcal{F}(C) \neq \emptyset$. There exists $K > 0$ such that

$$\|\mathcal{S}^n(u) - \mathcal{S}^n(v)\|_\infty \leq \frac{K^n}{n!}\|u - v\|_\infty \qquad \forall (u,v) \in \mathcal{F}(y)^2,\ \forall n \geq 0. \tag{9}$$

Furthermore, if $\mathcal{S}(\mathcal{F}(y)) \subset \mathcal{F}(C)$, then $\mathcal{S}$ admits a unique fixed point $u_* \in \mathcal{F}(y)$, and the Picard iteration $u^{(n+1)} = \mathcal{S}(u^{(n)})$ converges to u_* for any $u^{(0)} \in \mathcal{F}(y)$.

1. Two key assumptions: $\mathcal{F}(C) \neq \emptyset$ and self-mapping condition $\mathcal{S}(\mathcal{F}(y)) \subset \mathcal{F}(C)$;
2. Consequence of the estimate

$$|\mathcal{S}(u)(t) - \mathcal{S}(v)(t)| \leq \mathcal{V}|u - v|(t) \qquad \forall t \in [0,T],\ \forall (u,v) \in \mathcal{F}(y)^2 \tag{10}$$

where $\mathcal{V} : L^\infty \to L^\infty$ is a Volterra-type operator, i.e., for some $K > 0$, it holds $\|\mathcal{V}^n\| \leq \frac{K^n}{n!}$, $\forall n \geq 0$.

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Steering control and necessary condition for optimality

Theorem 3 [T, Ching; '25]

Let $(x^0, x^1) \in \mathbb{R}^d \times \mathbb{R}^d$ and set $y := x^1 - \phi_T(x^0)$. Assume $\mathcal{F}(C) \neq \emptyset$ and $\mathcal{S}(\mathcal{F}(y)) \subset \mathcal{F}(C)$ is satisfied. Let $\bar{u} \in \mathcal{F}(y)$ be the fixed point of $\mathcal{S}$ given by

$$\bar{u}(t) = B(x_{\bar{u}}(t))^\top D\phi_{T-t}(x_{\bar{u}}(t))^\top \mathcal{N}(\bar{u})^{-1} y. \tag{11}$$

Then $\bar{u}$ steers (Σ) from x^0 to x^1 on $[0, T]$. Furthermore, one has the energy identity

$$\|\bar{u}\|_{L^2}^2 = y^\top \mathcal{N}(\bar{u})^{-1} y. \tag{12}$$

Moreover, if $\bar{u}$ is a local minimum of $\frac{1}{2}\|u\|_{L^2}^2$ over $\mathfrak{F} := \{u \in \mathcal{F}(y) : F(u) := \mathcal{L}_u u - y = 0\}$, then

$$\langle [\mathcal{L}_{\bar{u}} DF(\bar{u})^*]^{-1} y, DF(\bar{u})\, h\rangle_{\mathbb{R}^d} = 0 \qquad \forall h \in \ker \mathcal{L}_{\bar{u}}. \tag{13}$$

- Fundamentally a synthesis result: given the fixed point $\bar{u} \in \mathcal{F}(y)$ of $\mathcal{S}$, it provides an explicit representation of a control that steers x^0 to x^1;
- Reachability depends on the structure of $\mathcal{F}(C)$ and on whether the self-mapping $\mathcal{S}(\mathcal{F}(y)) \subset \mathcal{F}(C)$ is satisfied.

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Outline

Fully-actuated control-affine system with an invertible input matrix

Proposition 1 [T, Ching; '25]

Suppose $d = k$ and there exists $b > 0$ such that

$$\sigma_{\min}(B(x)) \;\geq\; b, \qquad \forall x \in \mathbb{R}^d. \tag{14}$$

Then $\mathcal{N}(u) := \mathcal{N}(u, x^0)$ is uniformly coercive,

$$z^\top \mathcal{N}(u)\, z \;\geq\; \frac{1 - e^{-2\Lambda_1 T}}{2\Lambda_1}\, b^2\, |z|^2, \qquad \forall (z,\, x^0,\, u) \in \mathbb{R}^d \times \mathbb{R}^d \times L^\infty. \tag{15}$$

1. **Our steering control:** $\bar{u}(t) = B(x_{\bar{u}}(t))^\top D\phi_{T-t}(x_{\bar{u}}(t))^\top \mathcal{N}(\bar{u})^{-1} y$ steers (Σ) from any $x^0 \in \mathbb{R}^d$ to any $x^1 \in \mathbb{R}^d$;
2. **Feeback linearization control:** $u_{fl}(t) = B(x(t))^{-1}\Big(\frac{x^1 - x^0}{T} - N(x(t))\Big)$ steers (Σ) from any $x^0 \in \mathbb{R}^d$ to any $x^1 \in \mathbb{R}^d$.

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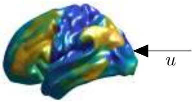
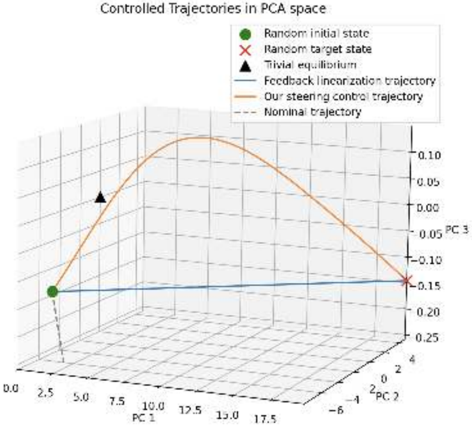
Fully-actuated 100D continuous-time Hopfield-type RNN

100D MINDy models [Chen *et al.*, '25]



$$\dot{x}_i = -d_i\, x_i + \sum_{j=1}^{100} w_{ij}\sigma_j(x_j) + \sum_{j=1}^{100} \delta_{ij} u_j, \qquad i = 1, \cdots, 100.$$

- Here, δ_{ij} is Kronecker's symbol;
- $d_i > 0$, and $w_{ij} \in \mathbb{R}$ are fitted on individual Brain dynamics using fMRI;
- $\sigma_j(x_j) = \sqrt{\alpha_j^2 + (bx_j + 0.5)} - \sqrt{\alpha_j^2 + (bx_j - 0.5)}$, where $\alpha_j > 0$ and $b > 0$ are also fitted.



$T = 1$, $\|\bar{u}\|_2 = 23.057$, $\|\bar{u}\|_\infty = 4.67$, $|x_{\bar{u}}(T) - x^1| = 10^{-11}$;
$\|u_{fl}\|_2 = 23.049$, $\|u_{fl}\|_\infty = 4.61$, $|x_{u_{fl}}(T) - x^1| = 10^{-13}$.

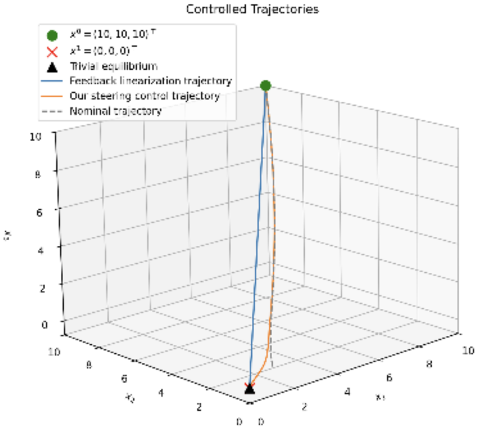
Fully-actuated 3D continuous-time Hopfield-type RNN

A 3D toy example

$$\dot{x}_i = -d_i\, x_i + \sum_{j=1}^{3} w_{ij}\sigma_j(x_j) + \sum_{j=1}^{3} \delta_{ij} u_j, \qquad i = 1, 2, 3.$$

- Here, $d_1 = 1.25$, $d_2 = 1.5$, $d_3 = 1$;
- $\sigma_j(x_j) := (1 + e^{-x_j})^{-1}$, and

$$(w_{ij}) = \begin{pmatrix} 3 & 1 & -0.5 \\ 2 & 1 & 0.5 \\ 0 & -1.5 & 1.25 \end{pmatrix}.$$



$T = 5$, $\|\bar{u}\|_2 = 4.60$, $\|\bar{u}\|_\infty = 4.33$, $|x_{\bar{u}}(T) - x^1| = 10^{-11}$;
$\|u_{fl}\|_2 = 16.12$, $\|u_{fl}\|_\infty = 9.50$, $|x_{u_{fl}}(T) - x^1| = 10^{-14}$.

General sufficient conditions for the existence of our steering control

Proposition 2 [T, Ching; '25]

Let $x^0 \in \mathbb{R}^d$. Assume that $\exists u^\flat := u^\flat(x^0) \in L^\infty$ s.t. $\mathcal{N}(u^\flat) := \mathcal{N}(u^\flat, x^0)$ is invertible and

$$\|u^\flat\|_\infty < \frac{\lambda_{\min}(\mathcal{N}(u^\flat))}{2\, L_{\mathcal{N}}}. \tag{16}$$

Then, letting $C := 2/\lambda_{\min}(\mathcal{N}(u^\flat))$ yieilds $\mathcal{S}(\mathcal{F}(y)) \subset \mathcal{F}(C)$ for every $x^1 \in \mathbb{R}^d$ satisfying

$$|y| := |x^1 - \phi_T(x^0)| \leq \frac{\lambda_{\min}(\mathcal{N}(u^\flat))}{2\,\|B\|_\infty\, e^{\Lambda_1 T}} \left(\frac{\lambda_{\min}(\mathcal{N}(u^\flat))}{2\, L_{\mathcal{N}}} - \|u^\flat\|_\infty \right). \tag{17}$$

Here $L_{\mathcal{N}} > 0$ is the Lipschitz constant of the map $u \in \{u \in L^\infty : \|u\|_\infty \leq \zeta\} \mapsto \mathcal{N}(u)$.

1. **Our steering control:** $\bar{u}(t) = B(x_{\bar{u}}(t))^\top D\phi_{T-t}(x_{\bar{u}}(t))^\top \mathcal{N}(\bar{u})^{-1} y$ steers (Σ) from any $x^0 \in \mathbb{R}^d$ to any $x^1 \in \mathbb{R}^d$ whenever (17) is satisfied;
2. Other steering controls are available if (Σ) is **completely controllable**: homotopy continuation or nipoltent approximation [**Sussmann, Chitour, Jean, Trélat, Long, Schmoderer, Chen, Ji, Boizot, Gauthier, Prandi**,...]

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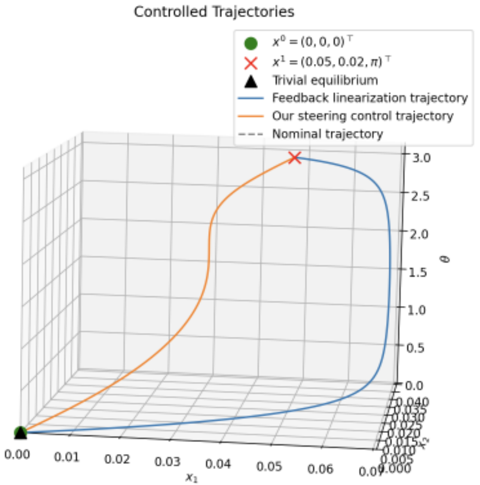
A 3D driftless system: the unicycle

Unicycle kinematic model

$$\dot{x} = \begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{\theta} \end{pmatrix} = \begin{bmatrix} \cos\theta & 0 \\ \sin\theta & 0 \\ 0 & 1 \end{bmatrix} \begin{pmatrix} u_x \\ u_\theta \end{pmatrix} =: B(x)u$$

- The unicycle is completely controllable on $[0, T]$ for any $T > 0$;
- The unicycle admits[a] a feedback linearization steering control u_{fl} from any $x^0 \in \mathbb{R}^2 \times \mathbb{S}^1$ to any $x^1 \in \mathbb{R}^2 \times \mathbb{S}^1$.

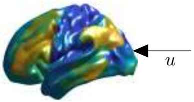
[a] Siciliano B., Sciavicco L., Villani L., and Oriolo G., Robotics: Modelling, Planning and Control, Springer, 2009.



$T = 0.15$, $\|\bar{u}\|_2 = 8.11$, $\|\bar{u}\|_\infty = 20.94$, $|x_{\bar{u}}(T) - x^1| = 10^{-15}$;
$\|u_{fl}\|_2 = 13.74$, $\|u_{fl}\|_\infty = 116.75$, $|x_{u_{fl}}(T) - x^1| = 10^{-15}$.

3D continuous-time Hopfield-type RNN with 2D input

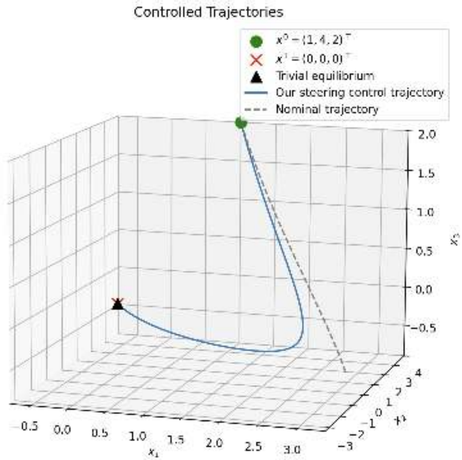
A 3D toy example



$$\dot{x}_i = -d_i\, x_i + \sum_{j=1}^{3} w_{ij}\sigma_j(x_j) + \sum_{j=1}^{2} \delta_{ij} u_j, \qquad i = 1, 2, 3.$$

- Here, $d_1 = 1.25$, $d_2 = 1.5$, $d_3 = 1$;
- $\sigma_j(x_j) := (1 + e^{-x_j})^{-1}$, and

$$(w_{ij}) = \begin{pmatrix} 3 & 1 & -0.5 \\ 2 & 1 & 0.5 \\ 0 & -1.5 & 1.25 \end{pmatrix}.$$

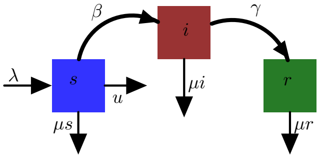


$T = 5$, $\|\bar{u}\|_2 = 6.57$, $\|\bar{u}\|_\infty = 5.16$, $|x_{\bar{u}}(T) - x^1| = 10^{-11}$.

Remark: We do not know *a priori* that the system is completely controllable on $[0, 5]$.

3D SIR model with 1D input

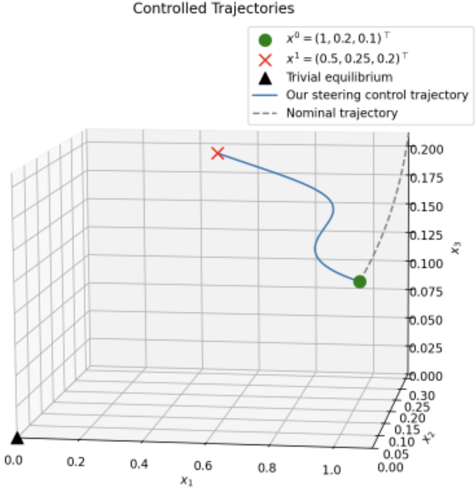
A 3D SIR toy example



Controlled SIR epidemic model

$$\dot{s} = \lambda - \beta si - \mu s - us,$$
$$\dot{i} = \beta si - (\gamma + \mu)i, \quad \dot{r} = \gamma i - \mu r.$$

For simulations, we use $\lambda = 1$, $\beta = 2$, $\mu = 0.2$ and $\gamma = 1$.



$T = 0.5$, $\|\bar{u}\|_2 = 1.91$, $\|\bar{u}\|_\infty = 6$, $|x_{\bar{u}}(T) - x^1| = 10^{-11}$.

Remark: We do not know *a priori* that the system is completely controllable on $[0, 0.5]$.

Planar control-affine systems with scalar inputs

Proposition 3 [T, Yu, Ching; '26]

Assume

$$\inf_{x\in\mathbb{R}^2} \left| \det \left(B,\ DN(x)B\right)\right| \geq \beta > 0. \tag{18}$$

Fix $x^0 \in \mathbb{R}^2$ and $U > 0$. Then there exists $\alpha_{\mathcal{N}} = \alpha_{\mathcal{N}}(\beta, |x^0|, U) > 0$ such that

$$\mathcal{N}(u) \succeq \alpha_{\mathcal{N}}\,\mathrm{Id}, \quad \forall u \in L^\infty(t_0, T) \text{ with } \|u\|_\infty \leq U.$$

1. Proposotion 3 is insightful merely when there is no control constraints *a priori*;
2. In this case, **our steering control** $\bar{u}(t) = B^\top D\phi_{T-t}(x_{\bar{u}}(t))^\top \mathcal{N}(\bar{u})^{-1}y$ steers (Σ) from any $x^0 \in \mathbb{R}^2$ to any $x^1 \in \mathbb{R}^2$ whenever (18) is satisfied.

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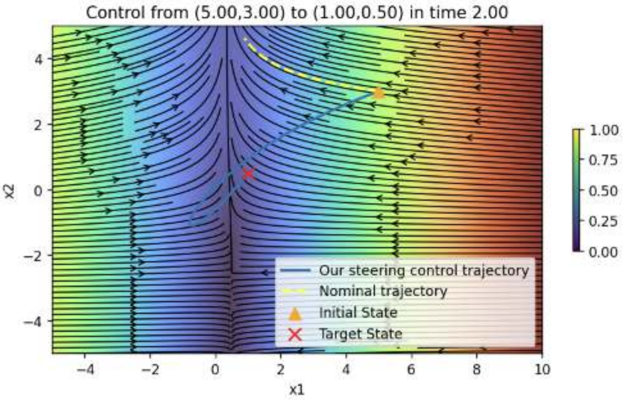
2D continuous-time Hopfield-type RNN with a scalar input

A 2D toy example

$$\dot{x}_i = -d_i\, x_i + \sum_{j=1}^{2} w_{ij}\sigma_j(x_j) + u, \qquad i = 1,\, 2.$$

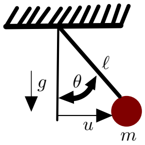
- $d_i > 0$, $w_{ij} \in \mathbb{R}$ and $\sigma_i(\cdot)$ is a bounded sigmoid;
- The system is completely controllable on $[0, T]$ for any $T > 0$ by **[Sun, '07]**;
- For simulations: $d_1 = 1.25$, $d_2 = 0.01$, $\sigma_j(x_j) = \tanh x_j$, $(w_{ij}) = \begin{pmatrix} 1 & -0.15 \\ -0.15 & 1 \end{pmatrix}$.



$\|\bar{u}\|_2 = 4.85$, $\|\bar{u}\|_\infty = 5.46$, $|x_{\bar{u}}(T) - x^1| = 10^{-11}$.

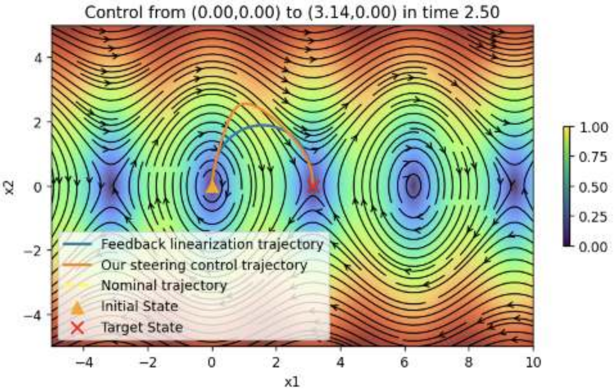
Torque-controlled pendulum (TCP)

Torque-controlled pendulum



$$\dot\theta = \omega, \quad \dot\omega = -a\sin\theta + bu$$
$$a := g/\ell, \quad b := 1/m\ell^2.$$

- TCP is completely controllable on $[0,T]$ for any $T>0$;
- $u_{fl}(t) = (v(t) + a\sin x_1(t))/b$ is a feedback linearization steering control from any $x^0 = (\theta^0, \omega^0) \in \mathbb{S}^1 \times \mathbb{R}$ to any $x^1 = (\theta^1, \omega^1) \in \mathbb{S}^1 \times \mathbb{R}$ on $[0,T]$. Here, $v(\cdot)$ is the Gramian-based control for $\dot\theta = \omega$, $\dot\omega = v$;
- For simulation: $g = 9.81$, $\ell = 4$ and $m = 1$.



$\|\bar u\|_2 = 63.68$, $\|\bar u\|_\infty = 96.94$, $|x_{\bar u}(T) - x^1| = 10^{-11}$;
$\|u_{fl}\|_2 = 57.69$, $\|u_{fl}\|_\infty = 48.25$, $|x_{u_{fl}}(T) - x^1| = 10^{-13}$.

Outline

Takeaways and Perspectives

Takeaways

- **Goal:** explicit steering controls for control-affine systems $\dot{x} = N(x) + B(x)u$ (endpoint control with low L^2 energy).
- **Key idea:** a *flow-conjugate* solution representation yields a nonlinear Gramian $\mathcal{N}(u)$ (LTI Gramian as a special case).
- **Synthesis:** define a map $\mathcal{S}$ whose fixed points are steering controls; Picard iterations converge (**with a factorial decay**) under a coercivity/self-mapping regime.
- **Energy:** the fixed point satisfies $\|\bar{u}\|_{L^2}^2 = y^\top \mathcal{N}(\bar{u})^{-1} y$, and an orthogonality condition for local minimizers is provided.
- **Numerics:** validated on nonlinear benchmarks (pendulum, unicycle, SIR model, Hopfield-type networks), with clear gains vs. naive/Feedback-linearization controls.

Perspectives

- Understand when *complete controllability* can be effectively combined with the present framework to guarantee well-definiteness of our steering control.
- Develop *coercivity criteria* in higher-dimensional settings, guaranteeing well-definiteness of our steering control.
- Characterize nonlinear control-affine systems for which our steering control satisfies the *orthogonality condition*, guaranteeing a local minimizer.

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- **Synthesis:** define a map $\mathcal{S}$ whose fixed points are steering controls; Picard iterations converge (**with a factorial decay**) under a coercivity/self-mapping regime.
- **Energy:** the fixed point satisfies $\|\bar{u}\|_{L^2}^2 = y^\top \mathcal{N}(\bar{u})^{-1} y$, and an orthogonality condition for local minimizers is provided.
- **Numerics:** validated on nonlinear benchmarks (pendulum, unicycle, SIR model, Hopfield-type networks), with clear gains vs. naive/Feedback-linearization controls.

Perspectives

- Understand when *complete controllability* can be effectively combined with the present framework to guarantee well-definiteness of our steering control.
- Develop *coercivity criteria* in higher-dimensional settings, guaranteeing well-definiteness of our steering control.
- Characterize nonlinear control-affine systems for which our steering control satisfies the *orthogonality condition*, guaranteeing a local minimizer.

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Thank You!

Questions?

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